

KIRKLAND LAKE, ON, Canada

WMO: 713010

Lat: 48.151N

Lon: 80.012W

Elev: 1063

StdP: 14.14

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-22.7	-33.5	0.9	-28.7	-27.3	1.2	-22.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.1	68.1	80.7	65.8	77.6	64.4	71.4	79.1	69.4	75.5	67.5	73.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	111.3	74.3	67.3	104.4	72.4	65.5	97.8	71.0	35.9	79.5	34.2	75.3	32.6	73.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-37.3	89.4	5.8	3.0	-41.4	91.6	-44.8	93.4	-48.0	95.1	-52.2	97.3		
(5)			-37.4	74.6	5.8	2.5	-41.6	76.3	-45.0	77.8	-48.2	79.2	-52.4	81.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	4.1	6.4	18.9	34.0	50.0	60.4	65.5	62.8	55.5	41.9	27.8	12.5		
	DBStd	23.99	15.00	13.23	13.64	10.28	9.22	7.38	5.93	6.11	8.30	8.73	10.44	13.03		
	HDD50	6372	1422	1220	966	487	116	8	0	1	37	285	667	1163		
	HDD65	10478	1887	1640	1429	931	472	173	67	115	302	718	1116	1628		
	CDD50	1559	0	0	2	6	117	321	480	398	201	33	1	0		
	CDD65	189	0	0	0	0	8	36	81	48	16	0	0	0		
	CDH74	1588	0	0	1	2	95	370	683	324	108	5	0	0		
	CDH80	359	0	0	0	0	23	83	172	63	18	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	32.1	1.8	1.5	1.5	2.2	3.0	3.3	3.7	3.5	3.8	3.3	2.5	2.0		
	PrecMax	40.4	4.5	2.9	3.0	4.1	4.7	5.7	6.2	5.8	6.3	5.6	4.7	3.4		
(15) Precipitation	PrecMin	21.5	0.2	0.2	0.1	0.7	1.5	2.0	2.0	1.5	1.6	1.3	1.2	0.7		
	PrecStd	3.7	0.9	0.8	0.7	0.9	0.8	0.9	1.2	1.1	1.0	1.1	0.8	0.7		
	DB	38.9	40.0	57.3	68.3	83.0	86.9	89.1	85.3	82.1	71.7	55.7	41.4			
	MCWB	38.1	37.3	47.7	53.3	63.7	70.1	69.5	69.6	69.5	61.8	51.8	40.7			
	2% DB	34.5	34.2	48.0	60.6	75.7	82.1	84.6	81.4	76.9	65.9	49.2	36.4			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.0	32.4	39.7	46.8	59.0	65.9	68.0	67.5	66.3	57.8	46.0	35.5			
	5% DB	31.5	30.3	43.0	55.2	71.2	78.4	81.0	77.5	73.2	60.4	45.4	33.4			
	MCWB	30.8	28.4	37.3	44.9	57.1	63.5	66.3	65.7	63.7	54.8	42.8	32.8			
	10% DB	27.7	26.8	38.5	50.1	66.6	74.3	77.6	74.2	69.2	55.6	41.5	31.3			
	MCWB	27.1	25.6	34.1	42.2	54.5	61.9	64.8	64.3	62.9	51.6	39.7	30.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	38.6	37.4	52.8	56.9	69.3	73.5	74.7	73.1	71.4	63.5	52.2	40.7			
	MCDB	38.8	39.4	55.0	66.3	77.7	83.0	84.6	80.7	79.4	69.0	54.9	40.9			
	2% WB	34.2	32.6	42.1	49.9	63.7	70.1	71.4	70.0	68.7	59.2	46.8	35.3			
	MCDB	34.4	34.3	46.7	56.4	70.0	76.6	78.4	76.6	73.9	64.7	48.3	35.8			
	5% WB	30.9	28.9	37.9	46.1	60.1	67.2	69.4	68.1	66.2	55.9	43.0	32.9			
(31) Mean Daily Temperature Range	MCDB	31.4	30.0	41.8	53.3	67.8	73.1	75.7	73.8	71.1	60.0	44.8	33.3			
	10% WB	27.0	25.5	34.4	42.6	56.2	64.4	67.3	66.2	63.4	51.9	39.6	30.3			
	MCDB	27.6	26.6	37.7	49.5	64.0	71.2	73.8	71.7	68.2	55.7	41.5	30.8			
	5% DB	20.3	23.2	24.3	20.6	21.1	19.8	19.0	17.6	17.4	13.2	12.1	16.0			
	MCDBR	14.2	19.0	25.1	29.1	27.6	25.6	23.4	22.0	22.6	21.4	16.4	12.6			
(36) Clear-Sky Solar Irradiance	MCWBR	13.7	16.9	18.1	17.2	13.5	12.1	10.1	10.5	12.3	13.7	13.3	12.3			
	5% WB	14.2	17.8	21.5	25.2	21.8	20.8	19.2	18.0	18.5	19.1	15.1	12.8			
	MCWBR	13.9	16.5	16.8	17.0	13.3	12.1	10.4	9.9	11.4	13.9	13.4	12.5			
	taub	0.264	0.275	0.307	0.339	0.355	0.374	0.389	0.380	0.358	0.322	0.288	0.255			
	taud	2.272	2.243	2.246	2.353	2.415	2.379	2.368	2.386	2.433	2.522	2.494	2.370			
(42) All-Sky Solar Radiation	Ebn at Noon	257	281	288	289	288	282	275	273	269	264	251	249			
	Edh at Noon	23	30	36	36	35	37	37	35	30	23	19	19			
	RadAvg	399	738	1199	1486	1660	1792	1733	1454	1093	642	340	271			
(45)	RadStd	63	106	123	129	139	116	145	95	97	87	36	35			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-2.09	-2.52	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page